

Roll No.....

MID SEMESTER EXAMINATION SEPTEMBER 2025

Name of the Program: **B.Tech. ME (AME)**

Semester: 5th

Name of the Course: **Automotives Fuels & Lubricants**

Course Code: TME514

Time: 90 Minutes

Maximum Marks: 50

Note:

- (i) Answer **all the questions** by choosing *any one of the sub questions*.
- (ii) Each question carries 10 marks

Q1	(10 marks)	CO1
(a)	Define and classify automotive fuels. Briefly explain the properties of gasoline and diesel with suitable examples.	
OR		
(b)	Differentiate between Octane Number and Cetane Number with their significance in SI and CI engines.	
Q2	(10 marks)	CO1
(a)	Explain the role of additives in improving the quality of gasoline and diesel fuels.	
OR		
(b)	Discuss the blending techniques used in reformulated gasoline and diesel to meet BS-VI fuel quality standards.	
Q3	(10 marks)	CO2
(a)	Compare the properties, advantages, and limitations of Ethanol and Methanol as automotive fuels.	
OR		
(b)	Explain the modifications required in IC engines when operating with alcohol-based fuels.	
Q4	(10 marks)	CO2
(a)	Write short notes on LPG, CNG, and Hydrogen as alternative automotive fuels.	
OR		
(b)	Discuss the safety and storage considerations of Hydrogen fuel in vehicles.	
Q5	(10 marks)	CO3
(a)	Describe the different testing methods used to evaluate the quality of gasoline and diesel fuels.	
OR		
(b)	Compare the performance and emission characteristics of biodiesel with conventional diesel fuel.	

